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Application of a new psychological assessment and training system in crew emergency simulation training

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***Abstract*—Shipboard personnel will inevitably face a series of unexpected problems when operating at sea, and in the actual maritime work, not only skilled operating techniques is needed, but also a strong psychological quality is also a crucial part when facing unexpected situations. In this paper, on the basis of the existing maritime operation training system, the psychological state assessment and training system for training personnel on board is added. By embedding hardware and software, the psychological state of personnel on board is detected in real time during the whole simulation training process, an accurate assessment of the psychological state is given, and psychological training is carried out for the training personnel's stress psychology and psychological load. This system can effectively improve the psychological quality of the shipboard personnel and provide them with the psychological quality to deal with the problems stably when they face the sea operation and unexpected situations in the future.**

Keywords-psychological training; psychological assessment; maritime emergency response; technology application;

I. Introduction

As the main tool for maritime operations, ships have an important position in maritime work[1]. It inevitably faces a variety of emergencies when operating at sea[2][3], in order to effectively strengthen the crew's ability to cope with such situations and improve the comprehensive quality of the crew, the crew emergency operation training system has been more common in the daily training[4]-[6]. Daily emergency simulation training, although it can to some extent improve the crew for fire, rescue and other emergency response capabilities, but it ignores the test of the psychological quality of the training personnel, individuals in the face of all kinds of emergency situations, psychological stress and psychological load is too high mainly in the mental state of the phenomenon of instability, which will lead to slower reaction, action deformation[7], affecting the emergency command and decision-making and emergency response ability to play; And excellent psychological quality can play an important role in solving all kinds of emergencies and avoiding disasters[8]. Therefore, this paper is based on the existing emergency operation training system, the introduction of training personnel psychological state assessment and training system, through the data collection terminal and evaluation software, etc., the physiological and behavioural indicators for the collection and analysis, the establishment of a statistical model will simulate the key parameters of training scenarios and the psychological state of the training personnel associated with the training effect assessment to provide a reference basis for the evaluation of the training effect of the feedback to adjust the difficulty of the emergency response scenario recommendations, to support the optimization of the content of emergency training for crew members. Support the optimisation of the crew's emergency training content.

II. Main Functions of the Psychological Assessment and Training System

The psychological assessment and training system is embedded in the crew's daily training process, using information collection terminals and system software to assess and train the trainer's state of mind, the system architecture is shown in Figure 1:

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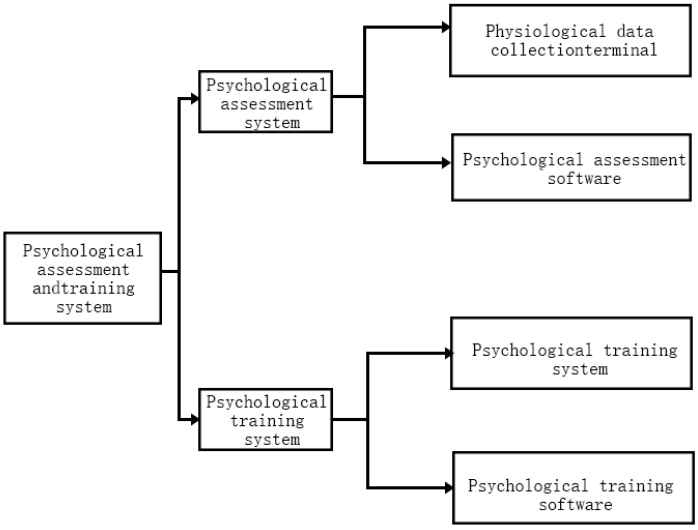

Figure 1. Architecture diagram of psychological assessment and training system.

As can be seen from the figure, the mental state assessment and training system is mainly divided into two parts: mental state assessment and mental training. The mental state assessment is mainly carried out by the information collection terminal and assessment software for real-time monitoring and assessment and analysis of the crew's training process, while the training system is mainly targeted at the mental stress and mental load assessed by the crew's training:

The psychological stress of the crew during training is mainly divided into three categories: tension, anxiety and emotional fluctuations[9]. Psychological training system for these three types of psychological stress to make targeted training: (1) through the emotional calming training to help the crew to establish a positive image for relieving tension. (2) Emotional activation training to help crew members remember the motivation to work in order to cope with anxiety and other avoidant emotions. (3) Relaxation training helps crew members to acquire relaxation skills such as breathing and meditation to stabilise their emotions. Through the targeted psychological training, the crew can transfer and apply the mastery of psychological soothing means to the emergency response, so that they can timely and effectively fight against psychological stress and maintain a stable psychological state in the emergency response scenario, and then devote all the cognitive resources to the operation to improve the level of emergency response.

On this basis, for the crew to deal with the psychological load of the unexpected situation, the system provides three core cognitive ability training means to enhance the alertness, relieve fatigue, and improve the concentration of attention during training: through perceptual training to help train the crew to maintain a higher sensitivity to the information in the environment to enhance the alertness, through the operation of the training to enhance the crew's ability to process information to combat fatigue, and through the attention training to improve the concentration of attention of the crew. The attention training can help the crew to maintain a high sensitivity to information in the environment, thus improving alertness. The core cognitive ability training can enhance the basic cognitive resources of the crew, so as to have a good reserve of resources to cope with the higher task load in emergency response, maintain a stable psychological state, and ensure the effective performance of emergency response skills.

III. Psychological Assessment and Training System Design Ideas

The new psychological state assessment system is based on the whole training system, using the embedded way through the whole training process, through the information acquisition terminal and psychological assessment software and algorithms to achieve the evaluation of the mental state of the trainers in the whole simulation training process. The specific design idea is shown in Figure 2:

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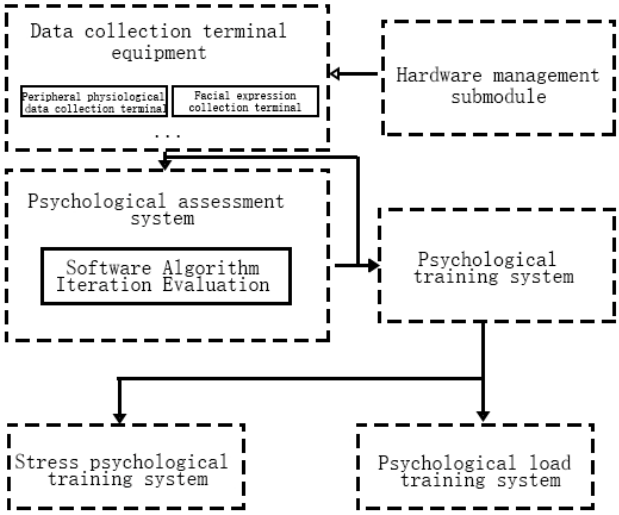

Figure 2. Development Path of Psychological State Assessment and Training Subsystem.

(1) In terms of hardware, in order to achieve the collection of multi-source and multi-modal physiological and behavioural data of crew members during the training process, the data collection terminal equipment is integrated with the simulator or virtual training equipment, and is bound to each data collection terminal equipment through the embedded way, and the hardware management sub-module is developed to drive the operation of the data collection terminal equipment, so as to ensure that the psychological state of the crew members in the real training can be collected.

(2) In terms of software algorithms, in order to achieve that the data preprocessing algorithm and the psychological state assessment algorithm can be applied to the real simulator training environment for crew members, we constructed a multimodal sample library of crew members' handling of the emergency requisition light in the training mode. By building a simulation test environment and conducting simulation test experiments, the high psychological stress and high mental load state of the training crew is induced, and the physiological and behavioural indexes during training are collected, which are used for the construction and optimization of the data preprocessing algorithm and the psychological state assessment algorithm.

(3) In realising the function of correlation analysis and psychological training, in order to realise the function of assessment and correlation analysis of the crew's psychological state, the sub-module of psychological state assessment and the sub-module of review are developed, and the synergy between the modules is realised. In order to realise the function of targeted psychological training for the crew, based on the psychological state during training obtained from the assessment, the stress psychological training sub-module and the load psychological training sub-module are developed, and psychological relief training is carried out for the crew with high stress level through the virtual scenario; and the core cognitive ability training is carried out for the crew through the portable terminal.

Sudden changes in the simulation training environment (e.g., training assessment, equipment failure, unskilled operation of the crew and unannounced inspections by the relevant departments, etc.) can lead to nervousness and anxiety among the trainers in response to the unfamiliar or sudden changes in the environment, which in turn generates psychological stress that affects the performance of skills[10]. Emergency situations in the crew simulation training system (e.g., hydraulic failure of the whole ship, rudder jamming of the lifting rudder, large longitudinal inclination, negative buoyancy, falling depth and collision danger, etc.) and emergency situations in the training programme (e.g., failure of the main hydrophobic system, failure of the high-pressure gas system, failure of emergency blowing system, failure of the ventilation system, failure of the hydraulic system, failure of the fire-fighting system, etc.), will also lead to crew members' emotions in the continuous emergency operation. fluctuation and fatigue, etc., causing high psychological stress and mental load. The crew psychological evaluation and training subsystem supports the evaluation of psychological stress and mental load in the simulation training of ship manoeuvring skills and the simulation training of virtual emergency handling conditions. It also correlates and analyses the elements of the task events such as rudder degree, rate of depth loss, longitudinal inclination angle, type of system failure, etc., with the above state of the trainers, and correlates and analyses the factors such as personnel, equipment, etc., with the psychological stress level of the trainers, so as to make clear the changes in the psychological state of the trainers. The key factors in the simulation training

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environment that lead to changes in the psychological state of the trainers.

IV. System Construction

A. Hardware function construction

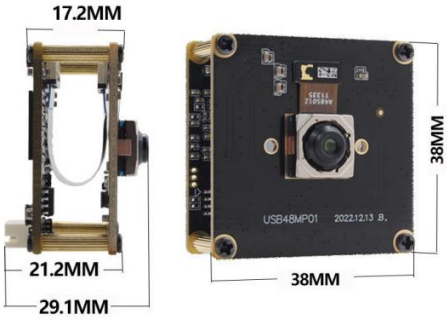
The information acquisition terminals include peripheral physiological data acquisition terminals, eye movement indicator acquisition terminals, facial expression capture terminals and voice information capture terminals. These terminals are responsible for collecting the physiological and behavioural indicators of the training crew. The peripheral physiological data collection terminal can collect physiological data such as heart rate, pulse, and electrocardiogram; the eye movement indicator collection terminal can capture eye movements and visual behaviour; the facial expression capture terminal can analyse changes in facial expressions to determine the state of emotions; and the voice information capture terminal can collect and analyse the emotions and contents of voices and perform voiceprint recognition.

During simulator training, some personnel do not have a fixed operating position and may move frequently during training, so it is suitable to use the peripheral physiological data acquisition terminal to collect their physiological data to assess the psychological state. The terminal has the advantages of being portable, lightweight and easy to wear, and it can effectively collect data by wearing it close to the skin one inch up from the left wrist of the ship's personnel, without interfering with the normal operation process of the trainees. As shown in Figure 3:

Figure 3. peripheral physiological data acquisition terminal.

The rest of the trainers are equipped with fixed manipulators for continuous monitoring and manipulation in a relatively fixed position, either wearing peripheral physiological data acquisition terminals or using cameras, eye-trackers and pick-ups in front of the electronic console to collect behavioural data such as facial expression information, eye-movement indicators, and training voice messages to assess psychological status. The camera, eye-tracker and pickup are non-contact assessment devices and have no impact on the training process. The eye movement indicators are collected with the crew's head facing the eye movement indicator collection terminal 60cm as the calibration position, and the indicators can be effectively collected within a range of 30cm from the front and back, 30cm from the left and right, and 27cm from the top and bottom of the calibration position; the facial expression information can be effectively collected within the horizontal rotation angle of the crew's face of ±30°, and within the pitch and tilt angle of ±15°; the voice information can be effectively collected within the range of 0-5m from the pickups; and the voice information can be effectively collected within the distance of the crew from the pickups. The voice information can be effectively collected within 0-5m from the pickup. Other terminals are shown in Figure 4:

(a) eye movement capture terminal



(b) Facial Expression Capture Terminal

(c) Voice Message Capture Terminal

Figure 4. Other Data Acquisition Terminals.

Through various physiological data acquisition terminals to collect all kinds of physiological data of the training crew during training, combined with psychological assessment software algorithms, iteratively analysed the psychological state of the training crew at this time, for the training of psychological stress and psychological load, through the psychological training equipment for further targeted training. Psychological training equipment includes virtual reality helmets for stress psychological training and computers and accessories for load psychological training. In terms of equipment performance, the virtual reality helmet presents a variety of immersive relaxation scenarios to the crew through two OLED screens with a frame rate of not less than 90Hz.The psychological training equipment is shown in Figure 5:

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Figure 5. Psychological training equipment.

B. Software function construction

As can be seen from Figure 1, the software is mainly for three functions: (1) Psychological state assessment function: the psychological state assessment function takes continuous key physiological and behavioural characteristics during the emergency training process as input, and outputs in real time the level of psychological stress (including nervousness, anxiety, and emotional fluctuations) and the level of psychological load (including alertness, fatigue, and concentration) of the crew members in the emergency scenarios. (2) Psychological training function: through the stress psychological training sub-module, it provides emotional calming training for the tension of psychological stress, emotional activation training for the anxiety of psychological stress, and relaxation training for the emotional fluctuation of psychological stress; through the load psychological training sub-module, it provides sensory training for the alertness of psychological load, executive function training for the fatigue of psychological load, and attention training for the attention concentration of psychological load. The load mental training sub-module provides perceptual training for the alertness of mental load, executive function training for the fatigue state of mental load, and attention training for the concentration of mental load. (3) Data management function: The data management function supports visual display, downloading, printing and exporting of training data and results of comparative analyses, as well as the generation of the crew's ability growth model and training reports. The functional design of the software evaluation and training software is shown in the figure 6:

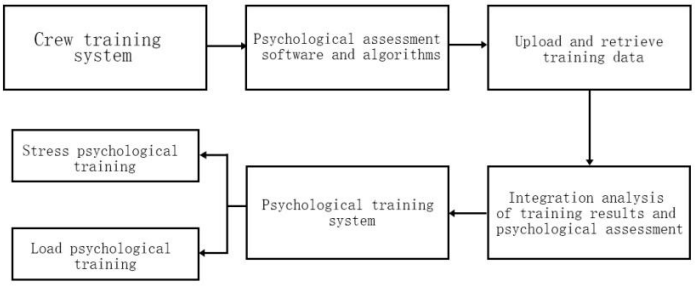

Figure 6. Functional design of psychological state assessment and training system software.

After a single training scenario, the system uploads physiological and behavioural monitoring data: the uploaded physiological and behavioural detection data are multimodal mental state fusion features. The different layers of the fusion features represent indicators from different sources, and the layers are peripheral physiological indicators, facial information, eye-movement indicators, and speech information, respectively. Since the features of each layer have different modalities, dimensions and weights, it is necessary to add positional information to the features of different layers when fused features are input into the model, as shown in Equation (1):

$$X_i = Input_{Embedding} + PE \tag{1}$$

Where Xi is the model input after adding the location information, which represents the features of the current layer, then it represents the current layer location encoding.

After getting the data input the self-attention mechanism as well as the multi-head attention mechanism. The calculation process will be input features after matrix operation to get q (Query,), k (Key,), v (Value), and then based on the above q, k, v for the self-attention mechanism operation, the process is shown in Equation (2):

$$Attention(Q,K,V) = Softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{2}$$

Where is the result of the operation of the self-attention mechanism, represents the vector dimension of k, and Softmax(x) represents the normalisation function. For any q, the weight of current attention is obtained by Softmax(x) activation function after calculating its dot product with k, and multiplied with v as the output of current locus.

Through the above algorithm combined with the data uploaded from the data collection terminal, the stress and load levels of the training crew and the training results are fused and analysed, and training recommendations are provided. Among them, for the training crew with high stress and high load, based on the results of the correlation analysis, the key task parameters that lead to the emergence of abnormal psychological state are clarified, and after targeted

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psychological training (including stress psychological training and load psychological training) is conducted, the above training process is repeated. By comparing and analysing the training results of multiple times for multiple people and groups, it generates the growth law model of the crew's emergency handling ability and psychological quality, and forms a reasonable and scientific parameter configuration scheme for emergency training subjects.

V. Conclusion

Currently psychological state assessment techniques based on physiological and behavioural multimodal data have been used on board driving simulators, flight simulators or in real flight and driving scenarios to assess the level of psychological state of drivers and pilots, etc. However, there are some differences between the crew's maritime contingency environment and the trained groups. The key technology in this paper is the psychological state assessment technology for trained crew members under simulated crew emergency scenarios. The technology adopts real-time data acquisition, collects and uploads the training data of the training crew through various hardware acquisition terminals (peripheral physiological acquisition terminal, eye movement acquisition terminal, facial expression acquisition terminal, etc.), and gives the psychological state of the crew through software algorithms, including psychological stress and psychological load, in which the psychological stress state is divided into nervousness, anxiety, and emotional fluctuations. And according to the different psychological state of the crew, with the help of psychological training system for further targeted psychological training of the crew, can strengthen the psychological quality of the crew and the ability to face the unexpected situation of stress. Therefore, this paper is of key significance for the future training of crew members in maritime operations and emergency situations.

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